



PUBLIC INCIDENT DIAGNOSIS

Home ISP Interactive Streaming Incident Diagnosis

Transport-path analysis • Avatel access

Date: 10 August 2026

Measurement window: 2026-08-10 10:55:56 to 11:09:31 UTC

Executive conclusion

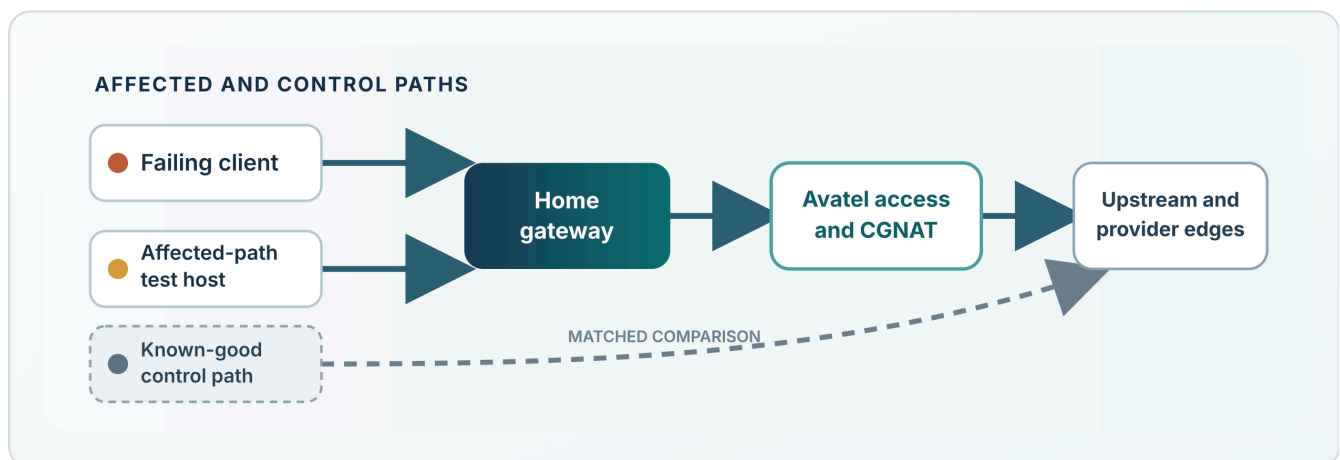
The highest-confidence cause is intermittent packet loss or short-lived blackholing on the direct home path, after the affected-path test host's clean Ethernet edge. The unresolved boundary includes the home router's WAN side, Avatel's access network and CGNAT, and upstream routing or peering. The evidence does not yet identify which member of that boundary drops the traffic.

The home router is **not excluded**. Clean Ethernet counters and reliable gateway replies prove only that one test host can reach the router over its local link. They say nothing decisive about the router's WAN interface, NAT table, firewall, flow offload, firmware, hardware, or upstream-facing link.

Generic direct-path impairment was reproduced: small provider connections sometimes timed out or retransmitted heavily, while large objects still completed at useful rates. The exact application failure was not reproduced under authentication: no authenticated Codex or Claude flow was captured on the actual failing client. The report therefore establishes a causal boundary, not a final component-level verdict.

Corrected topology and scope

The affected-path test host and the home router are separate machines. The test host sits behind the router and sends ordinary internet traffic through it. The router supplies ISP access through Avatel. Local bridge, forwarding, and DNS-service state observed on the test host (`br-lan`, `ip_forward`, and `dnsmasq`) is incidental; it does not make that host the household's ISP-access router and must not be used as evidence about the router's behavior.



The affected-path test host's public address was checked but is intentionally redacted. Ordinary IPv4 egress was confirmed to use Avatel. The known-good control path reached the same public services through a different uplink.

Representative measurement method

The investigation used read-only command shapes like the following on the affected-path test host and, where stated, the known-good control path. Targets, addresses, and resolver names are replaced with neutral placeholders.

```
# Route and egress proof
ip -4 route show default
ip rule show
curl -4 -sS --max-time <seconds> https://<egress-metadata-target>/<resource>

# Fresh HTTP/1.1, HTTP/2, and HTTP/3 timing batches
# <http-mode> was selected from --http1.1, --http2, and --http3-only
curl -4 -sS -o /dev/null <http-mode> --connect-timeout <seconds> \
  --max-time <seconds> \
  -w 'code=%{http_code} proto=%{http_version} connect=%{time_connect} tls=%{time_appconnect} first=%{time_starttransfer} total=%{time_total} bytes=%{size_download} error=%{error_msg}' \
  https://<target>/<resource>

# Fixed-address and persistent-connection variants
curl -4 --resolve '<service-name>:443:<address>' <timing-options> https://<service-name>/<resource>
curl <timing-options> -o /dev/null https://<target>/<resource> -o /dev/null https://<target>/<resource>

# Periodic drip with concurrent socket transport state
curl -4 -sS --no-buffer <http-mode> <timing-options> https://<neutral-drip-target>/<cadenced-resource>
ss -tin dst <resolved-target>

# TCP counter snapshots before and after a request batch
nstat -asz TcpRetransSegs TcpExtTCPTimeouts TcpExtTCPFastRetrans \
  TcpExtTCPMTUPFail TcpExtTCPMTUPSuccess

# Gateway, public-control, and TTL-limited probes
ping -4 -q -c <count> -i <interval> <gateway-or-control>
ping -4 -c 1 -W <seconds> -t <ttl> <control-target>

# Bulk and video-shaped transfer timing
curl -4 -sS -o /dev/null --max-time <seconds> \
  -w 'first=%{time_starttransfer} total=%{time_total} bytes=%{size_download} speed=%{speed_download}' \
  https://<bulk-target>/<size-resource>
```

Route and egress checks, fresh-request batches, persistent-connection batches, fixed-address batches, the HTTP-version matrix, periodic-drip/socket snapshots, counter snapshots, ICMP/TTL probes, and bulk transfers were separate batches. The two periodic-stream samples were matched by traffic shape and transport fields; they were not a simultaneous, clock-synchronized packet capture.

Observations

These statements come directly from measurements or reported incident history.

User-visible history

- Full-HD buffered video continued to work while Codex and Claude often stalled or delivered almost no interactive output.
- The same direct service had worked for about a month before the failure.
- No deliberate local configuration change was reported at onset.
- The home router was restarted repeatedly without durable improvement.
- An alternate VPN path made the interactive agents usable.

Local link and wide-area loss

- The home gateway answered without loss, at roughly 0.75 to 0.88 ms.
- Separate samples to two public anycast resolvers lost roughly 10 to 20 percent of probes.

- The affected-path test host's Ethernet interface showed no RX errors, TX errors, drops, carrier errors, or receive-queue drops in the inspected counters.
- During one short eight-request HTTP/2 batch, host TCP counters increased by 13 retransmitted segments, seven TCP timeouts, and four fast retransmits. Receive errors and receive-queue drops did not increase. These counters are host-global, so they corroborate impairment but do not attribute every event to the probe batch.

ICMP loss can be exaggerated by rate limiting. It is supporting evidence, not the sole basis for the diagnosis. TCP retransmission and application timing are the stronger witnesses.

Matched periodic-stream control

The same ten-second, one-byte-at-a-time stream was requested on the affected path and the known-good control path:

Measurement	Affected path	Known-good control
First byte	1.915750 s	0.280905 s
Total time	11.916481 s	10.396459 s
Retransmissions	5	0
Retransmitted bytes	5,066	0
Congestion window at observation	4	10
Socket rehash count	3	0
Path MTU	1500	1500
Advertised MSS	1448	1448
Observed MSS	1168	1168

Both streams completed, but only the affected path incurred retransmission, extra startup delay, a smaller observed congestion window, and socket rehashes. The equal PMTU and MSS values are strong evidence against a simple path-MTU mismatch explaining the difference. Neither path incremented `TCPMTUPFail` or `TCPMTUPSuccess` during the matched observation. Direct don't-fragment probing produced conflicting investigator accounts and is therefore classified as inconclusive. Congestion-window and rehash snapshots are transport symptoms, not proof of which network component caused them.

A second neutral streaming control delivered every expected byte for ten seconds over HTTP/1.1 and HTTP/2 on the affected path. Long-lived streaming is therefore not universally broken; the impairment is intermittent, path- or destination-sensitive, or both.

Fresh and persistent provider connections

- Fresh small HTTP/2 requests to one OpenAI address succeeded eight of eight times, while a second OpenAI address had three TCP-connect timeouts in eight attempts. The batches were sequential, so time-varying burst loss remains a confounder.
- A fixed Anthropic address had three TCP-connect timeouts in eight attempts on the affected path. The known-good control path completed twelve of twelve attempts to that address, typically in 0.11 to 0.29 seconds.

- Successful affected-path connections often showed retransmission-shaped timing steps: TCP establishment was normally about 12 ms but sometimes about 1.04 seconds, and TLS establishment ranged from roughly 46 ms to 1.6 seconds or more.
- On one already-established OpenAI HTTP/2 connection, a request received no bytes for ten seconds. Later requests then succeeded on that same connection in roughly 0.14 to 0.26 seconds.
- Unauthenticated OpenAI probes also produced two post-TLS no-response timeouts in eight attempts on the known-good control path. That endpoint behavior is a confounder and must not be presented as direct proof of the ISP fault.

Bulk and video-shaped controls

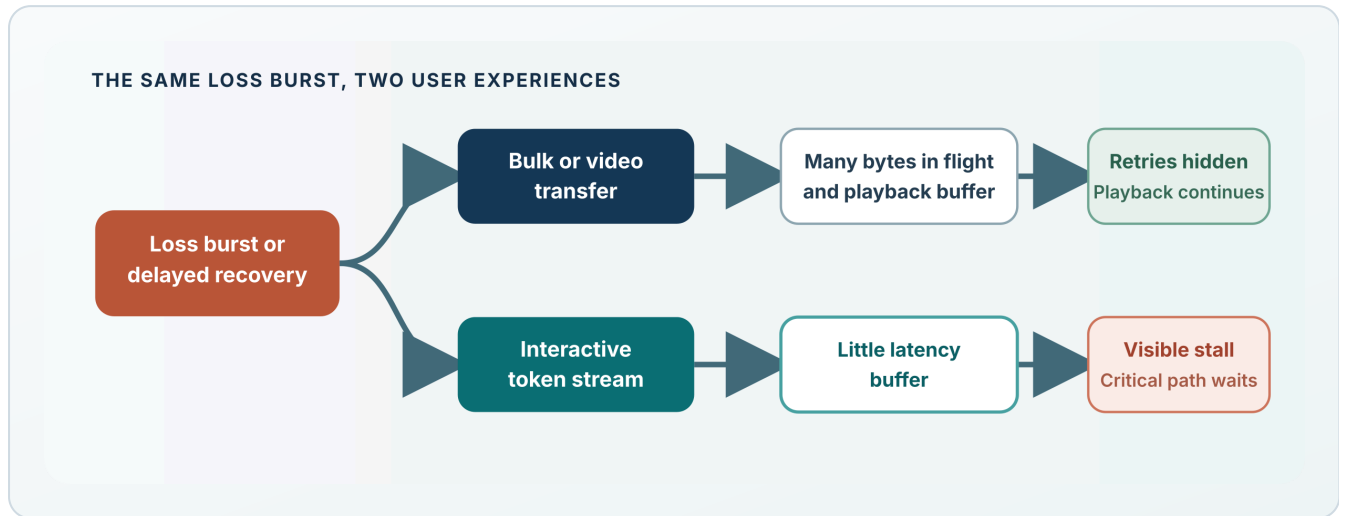
- A 1 MB Cloudflare object completed at 7.37 MB/s.
- A 10 MB Cloudflare object completed at 2.72 MB/s despite taking 3.39 seconds to deliver its first byte.
- An approximately 862 KB YouTube object completed despite taking 6.95 seconds to deliver its first byte.
- Three additional 5 MB bulk transfers completed at about 5.6 to 11 MB/s.

These measurements reproduce the apparent contradiction: useful aggregate throughput coexists with severe startup latency, retransmission, and occasional whole-request stalls.

Protocol controls

- DNS lookups were quick on the affected-path test host, and failures persisted there when provider addresses were fixed explicitly. DNS behavior on the actual failing client was not observed.
- Forced IPv4 reproduced failures. The affected-path test host had no ordinary global IPv6 route; forced IPv6 failed immediately rather than after a long stall.
- TLS completed successfully many times with HTTP/1.1 and HTTP/2 ALPN. Some failures occurred before TCP establishment, below TLS and HTTP.
- HTTP/3 succeeded from the affected path to neutral Cloudflare, Claude, and YouTube endpoints. UDP port 443 is not generically blocked.
- A ChatGPT HTTP/3 probe failed on both the affected path and known-good control path, so it is not a discriminating witness for this incident.
- Direct don't-fragment probing had conflicting investigator accounts and is conservatively treated as inconclusive. Matched sockets reported PMTU 1500, advertised MSS 1448, and observed MSS 1168; MTU-failure and MTU-success counters stayed at zero; multi-megabyte TCP transfers completed; and some failures occurred during SYN establishment, before payload size mattered.

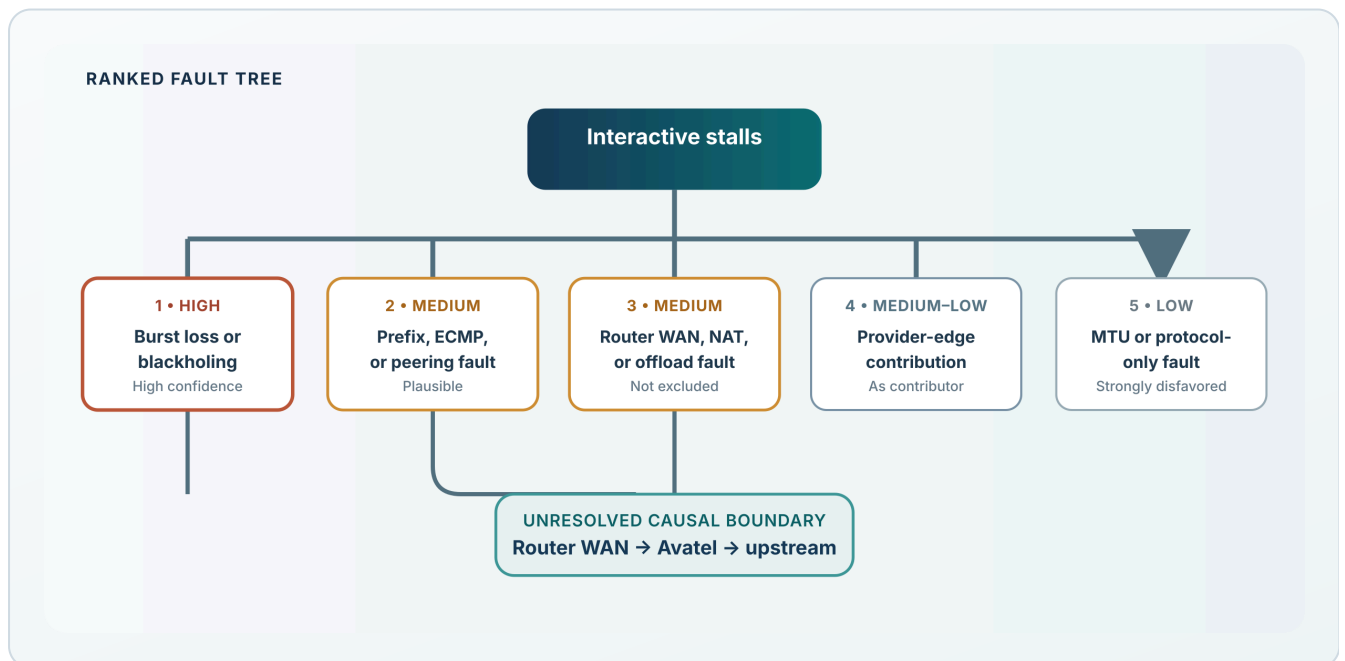
Why buffered video can work while interactive AI stalls



Video players prefetch, buffer, retry, and often maintain multiple opportunities to obtain the next media segment. A connection can pause for several seconds and still keep playing if the buffer is sufficiently full. Bulk transfer rates also average over the entire object, hiding slow first-byte time and short loss bursts.

Interactive agent output has a different contract. A handshake, request, or small response segment is on the user's critical path, and new tokens cannot be displayed until the missing transport data recovers. A three-second SYN loss, multi-second TLS retransmission, or ten-second response gap is immediately visible even if the connection later transfers data quickly. The measured bulk/stream contrast is consistent with this mechanism.

Ranked causes



1. **Intermittent packet loss or burst blackholing: high confidence.** The matched stream retransmitted only on the affected path; provider requests showed connection timeouts and retransmission-scale delays; wide-area loss appeared while the local gateway and Ethernet counters remained clean. The exact drop point is still unresolved.
2. **Destination-prefix, ECMP, routing, or peering impairment: medium confidence; plausible.** One OpenAI address performed materially worse than another, and the Anthropic direct-path result differed sharply from the known-good control. The address batches were sequential rather than interleaved, so a time-varying loss burst could mimic an address-specific fault. Unauthenticated OpenAI requests also timed out on the known-good control path, further weakening the provider-prefix inference.
3. **Persistent router WAN, NAT, firewall, hardware, firmware, or flow-offload behavior: medium confidence.** Router restarts and the absence of a reported configuration change make stale transient state or a newly introduced local rule less likely. A month of previous service also lowers the probability of a timeless configuration error. None of these facts excludes persistent WAN firmware, hardware, thermal, offload, NAT, or link behavior, nor a fault triggered by an ISP-side change.
4. **Provider-edge behavior: medium-low confidence as a contributor, low as the sole cause.** Unauthenticated OpenAI controls were imperfect even on the known-good path. However, the user-visible problem spans both OpenAI and Anthropic, and Anthropic showed a clean matched-path contrast.
5. **PMTU, DNS, IP-family, TLS, HTTP/2, or generic QUIC fault: low confidence as the primary cause.** Available controls strongly disfavor each simple version of these hypotheses, but do not rule them out on the actual failing client. They remain useful dimensions of the final authenticated trace.

Category disposition and missing witness

Category	Disposition	Evidence	One precise missing witness
Packet loss	Ruled in on the end-to-end direct path	Matched retransmissions, TCP timeout counters, provider connect failures, and wide-area loss	Simultaneous captures and interface counters at the failing client plus router LAN and WAN for the same five-tuple and timestamp
Reordering	Unknown; less supported than loss	Fast retransmits can follow loss or reordering; no per-flow SACK or DSACK trace exists	SACK, DSACK, and duplicate-ACK sequence from the same failed flow
MTU and PMTUD	Strongly disfavored as primary; not ruled out	Both matched paths reported PMTU 1500, advertised MSS 1448, and observed MSS 1168; TCPMTUPFail and TCPMTUPSuccess remained zero; large TCP objects succeeded; some failures occurred at SYN; direct DF accounts conflict	A clean DF size sweep plus packet capture for ICMP fragmentation-needed or packet-too-big messages while alternating default and 1200 MSS on the same authenticated flow
DNS	Strongly disfavored on the affected-path test host; actual failing-client DNS unknown	Test-host resolution was fast and failures persisted with fixed addresses	Scoped actual-client resolver trace and cache state plus the chosen A/AAAA and connection target during an authenticated stall, paired with a fixed-address retry
IPv4 versus IPv6	Ruled out as primary on the test host	Forced IPv4 reproduced; IPv6 lacked an ordinary route and failed immediately	The failing client's route table and address-family choice during the authenticated stall
TLS and ALPN	Ruled out as sole root	TLS, HTTP/1.1, and HTTP/2 negotiated successfully; failures also occurred before TCP connect	Packet trace showing the exact TLS record or ACK lost during an authenticated stall
HTTP/1.1 versus HTTP/2	Generic protocol defect ruled out	Neutral periodic streams completed under both versions; some failures occurred before ALPN	The same authenticated request forced over HTTP/1.1 and HTTP/2, if the provider permits both

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HTTP/3 and QUIC	Generic UDP/443 block ruled out	HTTP/3 worked to three independent public services; the ChatGPT result failed on both paths	Negotiated ALPN and scoped UDP/TCP capture from the real Codex or Claude client
Fresh connections	Affected	Multiple provider TCP-connect timeouts and RTO-shaped establishment delays	Alternating authenticated attempts to fixed provider destinations with timestamps and per-flow transport state
Long-lived sockets	Intermittently affected, not generically broken	One established HTTP/2 request stalled for ten seconds and recovered; neutral periodic streams completed	Authenticated stream capture over the failing client's real cadence and idle duration
NAT timeout	Disfavored for immediate failures; unknown at minute scale	Fresh connections fail; a persistent connection recovered; ten-second streams completed	Hold one provider connection idle for 30, 60, and 120 seconds while observing router conntrack state, then send data
Router firewall and offload	Unknown; router not excluded	Test-host Ethernet and gateway are clean, but router LAN/WAN forwarding was not observed	Simultaneous router LAN- and WAN-interface capture and counters for one failed flow
CGNAT	Unknown as the drop point	Avatel's access path adds a stateful boundary outside the home router; no ISP-side state was visible	Avatel-side flow logs or a provider-synchronized trace for the same five-tuple and timestamp
Route asymmetry	Unknown	No usable forward-and-reverse route pair was captured	A reverse/provider-side trace synchronized with the client-side flow
Upstream filtering or shaping	Possible, not proven	VPN success moves the symptom away from the direct path, but also changes route, egress identity, MTU, DNS, and encapsulation	Identical authenticated fixed-destination request direct and through the tunnel, with synchronized packet traces
Failing-client host or application	Unknown	The affected-path test host reproduced generic impairment but is not necessarily the failing client	ss transport state and a scoped capture from the actual failing client during one exact Codex or Claude stall

Confounders and claim limits

- No authenticated Codex or Claude request was captured on the failing client. Provider status endpoints and unauthenticated API responses approximate the traffic path but do not reproduce the full application contract.
- The affected-path test host is a valid ISP-path witness, not a substitute for the failing client's network namespace, socket state, application version, or local link.
- ICMP responders may rate-limit probes. Packet loss conclusions therefore rely on the combined TCP, timing, and matched-stream evidence.
- Direct don't-fragment probe accounts conflict. This report retains no positive full-size DF claim and treats direct DF probing as inconclusive.
- Fixed provider-address batches were sequential. A temporary loss burst could look like destination-specific routing. An alternating schedule is required.
- Host-global TCP counter changes can include unrelated traffic. Per-flow packet evidence is required for final attribution.
- VPN success is a strong boundary control but changes several variables at once: route, egress identity, DNS behavior, effective MTU, encapsulation, and sometimes transport protocol.
- Previous month-long success, no reported configuration change, and repeated restarts lower the likelihood of a simple static misconfiguration or stale transient table. They do not exclude persistent router firmware, hardware, WAN offload, line, CGNAT, ISP routing, or peering faults.

Smallest safe discriminating test

Capture one timestamped, authenticated Codex or Claude stall on the actual failing client while observing the same narrowly filtered flow on both the home router's LAN and WAN interfaces. Pair the captures with read-only socket state from the client.

The result separates the boundary directly:

- If the client emits no request data, investigate the client or application.
- If the router receives the flow on LAN but does not emit it on WAN, investigate router NAT, firewall, firmware, or offload.
- If the router emits SYNs or data and retransmits on WAN but receives no response, investigate Avatel, CGNAT, upstream routing, peering, or filtering.
- If the router receives return traffic on WAN but the client does not receive it on LAN, investigate the router or local network.
- If acknowledgements arrive out of order or carry SACK/DSACK evidence, separate reordering from true loss.

This capture is operationally read-only but privileged and therefore requires separate authorization. A subsequent direct-versus-tunnel A/B of the identical authenticated, fixed-destination request would sharpen the result, but enabling or changing a tunnel mutates path state and also requires separate explicit authorization and a rollback plan.

Actions taken

Diagnosis used read-only routing, interface, resolver, socket-counter, ICMP, TLS, HTTP, and transfer probes. No host, router, service, route, VPN, firewall, DNS, NAT, or offload state was changed. No privileged packet capture was made.